

Which of the following best exemplifies instinctive drift?

- ☐ A. Children are born with a biological predisposition to prefer sweet foods and to dislike bitter foods, which are more likely to be toxic. [10%]
- ☐ B. In humans, it is easier to condition a fear response to innately dangerous stimuli, such as snakes, than to harmless stimuli, such as rabbits. [7%]
- ✓ ☒ C. Animals trained to perform a specific behavior will often lose that behavior in favor of innate behaviors, even when reinforcement is present. [72%]
- ☐ D. Animals who become ill after one exposure to a noxious stimulus will quickly learn to avoid that stimulus in the future. [9%]

Correct

73% answered correctly

Explanation:



An **instinct** is an **innate, fixed** pattern of behavior that is more complex than a reflex, which is a simple response to a stimulus (eg, jerking hand away from hot stove). Instincts are not based on prior experience or learning. For example, newly hatched sea turtles instinctively know to move toward the ocean and swim.

Instinctive (or instinctual) **drift** describes an animal's **innate behaviors overshadowing a learned behavior**. Animals can often be trained using operant conditioning, whereby a desired behavior is reinforced with a reward. However, even when reinforcement is provided, trained animals will often revert to innate behaviors.

For example, researchers used food rewards to train pigs to pick up wooden coins and deposit them in a piggy bank. Over time, even though the pigs could be successfully trained to complete the behavior, the pigs began dropping the coins before reaching the piggy bank and pushing them along the ground with their snouts, a species-specific behavior known as rooting.

(Choice A) Children are biologically predisposed to prefer sweet over bitter tastes,

which is believed to confer an evolutionary advantage to babies who preferred more calorie-dense (sweet) foods and rejected foods more likely to be toxic (bitter). However, this describes an innate preference and is not an example of instinctive drift.

(Choice B) Most animals have an innate fear of their predators. Although human behavior is more complex than animal behavior, studies suggest that it is easier to condition humans to fear certain stimuli, specifically snakes, spiders, and rats. However, this describes a fear instinct in humans, which is not an example of instinctive drift.

(Choice D) Animals and humans will associate whatever they consumed (usually through sight or smell cues) with becoming ill and will avoid it in the future. This is known as a taste aversion, a powerful type of associative learning, and is not an example of instinctive drift.

Educational objective:

Instincts are innate, fixed behaviors. Instinctive drift describes an animal's reverting from a learned behavior to an innate behavior despite reinforcement. Instinctive drift often involves behaviors related to food or exploration.

Time spent:0 sec

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